

Received March 16, 2021, accepted April 25, 2021, date of publication May 5, 2021, date of current version May 20, 2021.

Digital Object Identifier 10.1109/ACCESS.2021.3077703

Comparison of Deep Learning Techniques for River Streamflow Forecasting

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This work was supported by the National Research Foundation of Korea (NRF) Grant by the Korea Government through the Ministry of Science and ICT (MSIT) under Grant 2020R1A2C1102758.

ABSTRACT Recently, deep learning (DL) models, especially those based on long short-term memory (LSTM), have demonstrated their superior ability in resolving sequential data problems. This study investigated the performance of six models that belong to the supervised learning category to evaluate the performance of DL models in terms of streamflow forecasting. They include a feed-forward neural network (FFNN), a convolutional neural network (CNN), and four LSTM-based models. Two standard models with just one hidden layer—LSTM and gated recurrent unit (GRU)—are used against two more complex models—the stacked LSTM (StackedLSTM) model and the Bidirectional LSTM (BiLSTM) model. The Red River basin—the largest river basin in the north of Vietnam—was adopted as a case study because of its geographic relevance since Hanoi city—the capital of Vietnam—is located downstream of the Red River. Besides, the input data of these models are the observed data at seven hydrological stations on the three main river branches of the Red River system. This study indicates that the four LSTM-based models exhibited considerably better performance and maintained stability than the FFNN and CNN models. However, the complexity of the StackedLSTM and BiLSTM models is not accompanied by performance improvement because the results of the comparison illustrate that their respective performance is not higher than the two standard models—LSTM and GRU. The findings of this study present that LSTM-based models can reach impressive forecasts even in the presence of upstream dams and reservoirs. For the streamflow-forecasting problem, the LSTM and GRU models with a simple architecture (one hidden layer) are sufficient to produce highly reliable forecasts while minimizing the computation time.

INDEX TERMS Bidirectional LSTM, deep learning, gated recurrent unit, long short-term memory, streamflow forecasting.

I. INTRODUCTION

Floods are one of the natural disasters that can cause substantial socioeconomic and human losses globally. According to an annual report on natural disaster issues conducted by the Centre for Research on the Epidemiology of Disasters (CRED) in 2019, flooding has affected more people than any other natural disaster in the 21st century [1]. The statistics of catastrophes occurring globally in 2019 indicated that although disasters caused by flooding are just ranked 2nd in terms of their impact on the number of people, accounting

for about 33% (after the storm disaster with a rate of 35%), this disaster is especially dangerous because it causes the most deadly catastrophe compared to other disasters, with a rate of about 43.5% of deaths [2]. Therefore, river streamflow forecasting or flood forecasting is one of the most challenging tasks in water resource management, and the findings of streamflow studies will be vital references for regulators in making decisions and policies on monitoring and prevention to mitigate damage. Moreover, owing to the inherently heterogeneous hydrological characteristics of the basins, common issues often faced in streamflow prediction studies might come from uncertainty, instability, or sometimes inadequacy of hydrological data [3]. At present, there

The associate editor coordinating the review of this manuscript and approving it for publication was Grigore Stamatescu.

are two prevalent approaches identified to handle the streamflow-forecasting issue.

The first approach is understood as the model-driven approach which is based on the physical modeling of hydrological processes such as rainfall–runoff. This approach is widely applied globally as a trustworthy solution in hydrological simulation and forecasting [4]. Despite its long tradition, various weaknesses, such as performance [5] or uncertainty analysis [4], have been identified in previous studies. Sometimes the physical parameters of the model are difficult to calibrate and estimate to match watershed characteristics [6]. In addition, hydrological models of this type generally tend to require a vast amount of data with various input data types, for instance, data on topographic, precipitation, discharge, and water levels [7]. This is also one of the disadvantages of the model-driven method because sometimes the collected data are inconsistent or unavailable at all sites, especially for developing countries. Another possible issue in the construction of rainfall–runoff models is the close dependence on precipitation data. However, precipitation in certain cases is not the leading cause of river flooding as downstream flooding can occur due to the flow regulation of large numbers of dams or reservoirs located upstream, especially for river basins spanning many countries.

Moreover, the other approach is the data-driven approach. Contrary to the model-driven approach, which depends on the physical modeling of the runoff process, the data-driven approach is based on ideas from statistical analysis of both linear and nonlinear relationships between input and output data without any additional assumptions [7]. This method is receiving increasing interest from scholars due to the reduction of physical assumptions of the rainfall–runoff process [3]. Owing to breakthrough advances in computational science over the last decade, artificial neural network (ANN)-derived models—in particular, machine learning (ML) and deep learning (DL)—have received the most interest among data-driven methods.

Concerning the ML method, Mosavi *et al.* [5] conducted a literature review of papers on applying ML methods for flood forecasting. This study indicated that various ML techniques had been proposed, and these techniques can be used alone or along with physical models to minimize training errors and enhance flood prediction efficiency. Generally, the studies mentioned in this paper depicted a good performance of ML in flood prediction problems corresponding to specific research objectives and particular areas. Despite the remarkable results, hydrological data have the characteristics of sequential data (time-series data), which most ML models do not emphasize exploitation [8]. This might be due to the complexity of building an ANN-based ML model when working with time-series data and the close relationship of observed data with the subsequently observed values [9], [10].

Concerning time-series data problems, DL models, especially those based on recurrent neural network (RNN), have demonstrated superiority in resolving complex tasks.

RNN is one of the typical architectures that belongs to the supervised learning category according to the classification of DL (based on representation learning) and its design to handle sequential data efficiently. Long short-term memory (LSTM) and gated recurrent unit (GRU) are two typical architectures in RNN-derived architectures due to their similarity in design, and both are considered special evolutionary forms of RNN [11], [12]. Several studies have employed LSTM and GRU models in hydrological forecasting studies [7], [13]–[17]. In addition, several complex DL models based on LSTM and GRU architectures, such as convolutional GRU [18], encoder–decoder-LSTM [19], wavelet-LSTM and convolutional LSTM [20], LSTM-attention optimizer [21], or entity-aware-LSTM [22], have also been proposed to improve the predictive performance of the models. A noticeable feature in most of the aforementioned studies is the complexity of DL models because the recommended models are often a combination of LSTM (or GRU) with one or more various architectures. In addition, vast data type requirements for these models can increase data management complexity—in particular, in real-time streamflow forecasting—because it is sometimes difficult to obtain the required amount of data at all observation stations in a short-time period.

For a comprehensive evaluation of the performance of LSTM-based models in terms of streamflow forecasting, the performance comparison of simple LSTM-based models, such as LSTM and GRU, with complex LSTM-based models has been performed in this study. Here, the term “simple” implies models that use just one hidden layer in their structure versus using “complex” to depict multiple hidden layers. The simplification of the model structure may save computing resources and improve computational performance. Accordingly, the stacked LSTM (StackedLSTM) model and the Bidirectional LSTM (BiLSTM) model are two complex models used against LSTM and GRU models (with just one hidden layer). Besides, this study is interested in two other neural network models that belong to the supervised learning category: the feed-forward neural network (FFNN) and convolutional neural network (CNN).

The Red River basin (area in the territory of Vietnam) is adopted as a case study area because it is the largest river basin in the north of Vietnam (ranked 2nd nationwide) and plays a vital role in the economic development of Vietnam. In addition, the Red River basin is formed on the basis of the combination of three main river branches, and Hanoi city—the capital of Vietnam—is located downstream of the Red River basin. Thus, the efficiency of the river-runoff forecasting process has always been challenging to regulators. Further, the presence of dams and reservoirs located upstream of the Red River system (the part outside Vietnam) could, in certain cases, be one of the potential causes of the disaster, such as flooding. Required data minimization and model-structure simplification could increase the computation speed while maintaining accurate predictability of DL models. Hence, the input data of the six employed DL models are

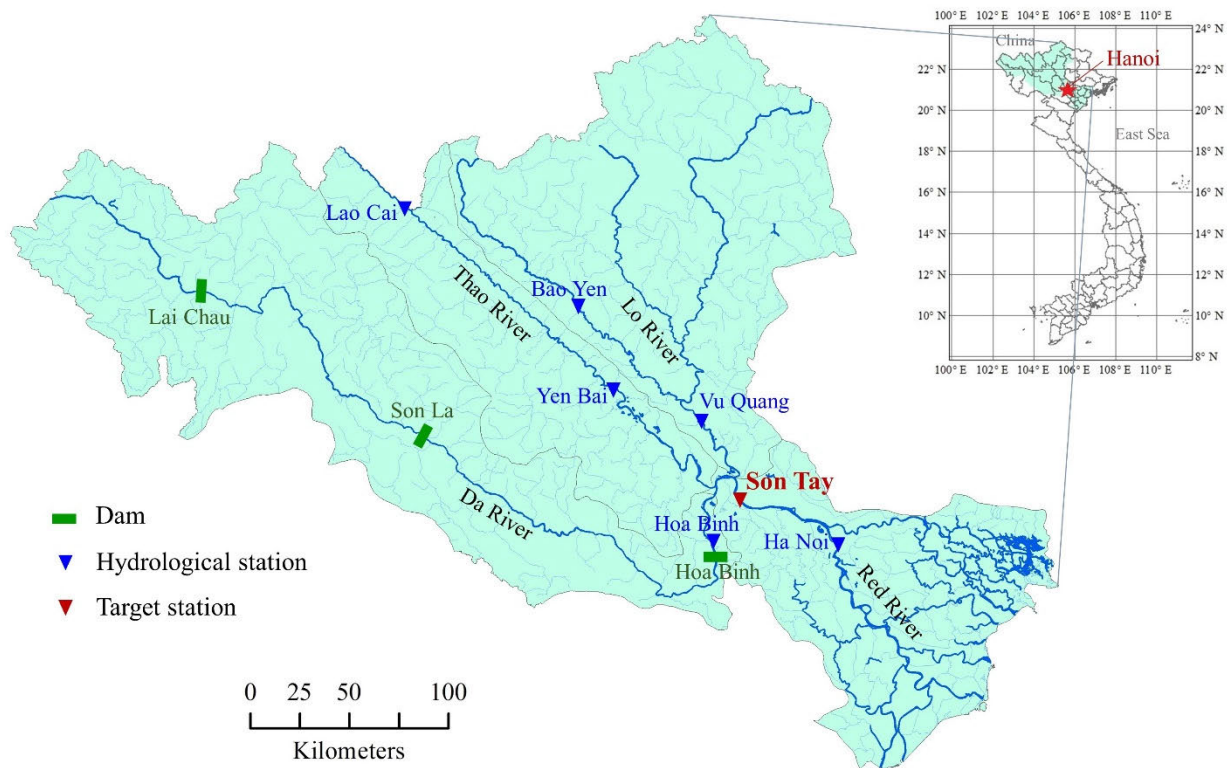


FIGURE 1. Location of hydrological stations in the Red River basin.

just data on observed discharge at the hydrological stations in the study area.

II. DATA AND METHODOLOGY

A. STUDY AREA AND DATA

The Red River basin is one of the largest river basins in Vietnam (ranked second after the Mekong River basin). Originating from China, it flows in a northwest–southeast direction into Vietnam and eventually discharges into the East Sea. The catchment area located in Vietnam is 87,840 km², accounting for about 51.3% of the entire basin area, of which the length of the Red River in the territory of Vietnam is about 556 km. The Red River basin system is formed from three major river branches, Da River, Lo River, and Thao River. Thao River is known as the mainstream of the Red River, with a catchment area of about 33.4%. The catchment areas corresponding to the Da River and Lo River are 33.7% and 24.5%, respectively. According to the flow monitoring data at hydrological stations, although the catchment areas of Thao River and Da River are approximate, the annual flow volume of Da River accounts for 45.2%, which is higher than that of Thao River (which is only about 21.8%). Lo River and the downstream area of the Red River account for 27.4% and 5.6% of the total annual flow of the Red River system, respectively. Fig. 1 shows a schematic of the study location.

For the hydrological characteristics, the seasons influence the flow in the Red River system. Accordingly, the amount of

flow in the rainy season (usually starting from late April to November) accounts for about 70%–80% of the total annual water volume [23]. Observed data at Son Tay hydrological station have recorded that the annual water level can vary from 5 to 8 m during the flood season, especially reaching 14 m during the typical flood in 2002 [8]. Besides, there are three large hydroelectric dams—Son La, Hoa Binh, and Lai Chau hydroelectric dams—on the Da River with a total installed capacity of 5,520 MW. The Son La hydroelectric dam (operated in 2012) has the highest power capacity of 2,400 MW, followed by Hoa Binh hydroelectric dam (operated in 1994) and Lai Chau hydroelectric dam (operated in 2016) with power capacities of 1,920 and 1200 MW, respectively. The effects of these dams on the flow of the Da River might be seen more clearly during the flood season when hydropower dams are involved in the flow regulation process. Owing to the complexity of the Red River system, the flood peaks identified on the main tributaries may also occur not at the same time. Moreover, the flood occurring on the Red River (observed data at Son Tay hydrological station) is often a combination of the flood of several major branches. A good example for this would be the case of the typical flood in 2002 (Fig. 2).

In this study, daily runoff data were collected from seven hydrological stations on the Red River system. Particularly, five hydrological stations—Lao Cai and Yen Bai stations in Thao River, Hoa Binh station in Da River, and Bao Yen

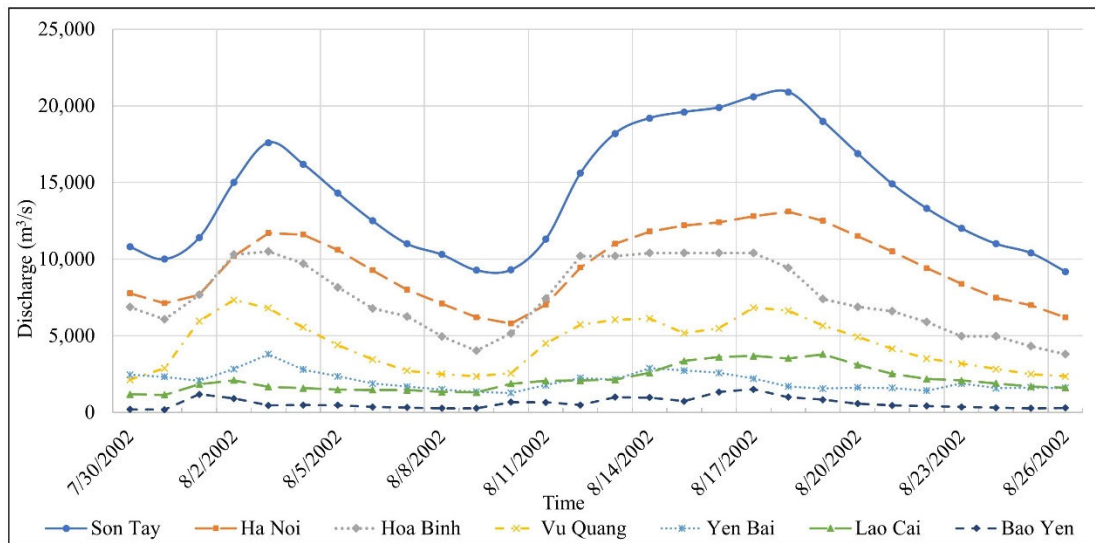


FIGURE 2. Observed data at hydrological stations during the typical flood recorded in 2002.

TABLE 1. Correlation matrix of the input data.

Stations	Bao Yen	Lao Cai	Yen Bai	Vu Quang	Hoa Binh	Ha Noi	Son Tay*
Bao Yen	1						
Lao Cai	0.65	1					
Yen Bai	0.69	0.88	1				
Vu Quang	0.7	0.74	0.76	1			
Hoa Binh	0.57	0.75	0.73	0.79	1		
Ha Noi	0.65	0.83	0.82	0.88	0.93	1	
Son Tay*	0.64	0.78	0.78	0.87	0.91	0.96	1

* denotes the target–forecast hydrological station.

station and Vu Quang station in Lo River—are located in the upstream area, one hydrological station is located downstream of the Red River—Hanoi station—and the target–forecast station—Son Tay station—is on the Red River. The Son Tay hydrological station is located in the northwest of Hanoi city—the capital of Vietnam. Therefore, the Son Tay station was selected as the case study because of the importance of its geographical position compared to other hydrological stations, such as the Hanoi hydrological station (Fig. 1). The findings of this study will serve as a precursor to developing an early flood-warning system at the Son Tay station in the Red River for Hanoi city. Table 1 provides information on the correlation matrix of the discharge data between the target and other stations.

The input data for this study were the daily data series on the flowrate recorded at the seven hydrological stations for 20 years from 1995 to 2014. Other data on precipitation, water levels, topographic features, and land use are not required

in this study. This is because input data type (data quality) has a more prominent effect on model performance than the quantity of data [7]. This means that runoff data will be collected in priority in the case of stream forecasting issues compared with other data, such as water levels or precipitation. In addition, simplifying the requirement for the amount of input data will contribute to minimizing the dependency on data collection because it is sometimes challenging to collect multiple types of data simultaneously and at all sites.

The available dataset was divided into two sections for different purposes. The first section was a series of observed flowrate data of 19 years (1995–2013), which were used for training and validation purposes. Particularly, 80% dataset was exploited for training and model evaluation, and the remaining 20% dataset was used for validating and selecting the best parameters for the models. The second section (collected dataset in 2014) was used as an unseen dataset to evaluate the model performance with two forecast cases—one day and two days ahead at Son Tay hydrological station. These data were measured in cubic meters per second (m^3/s).

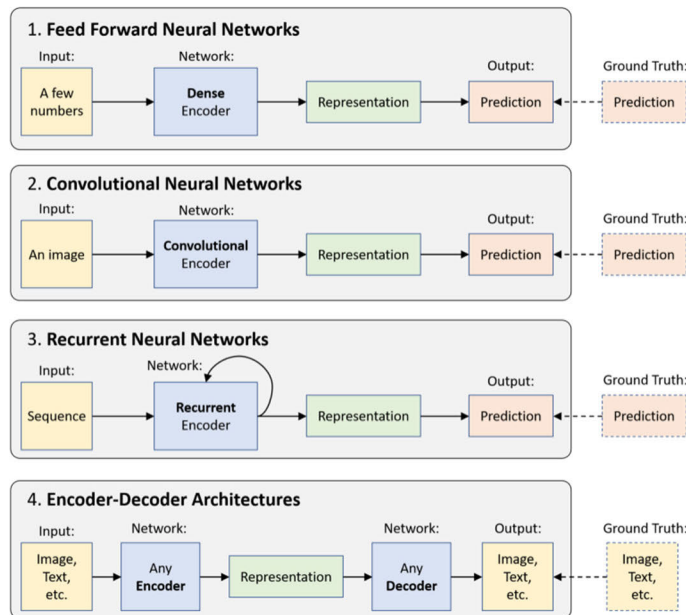
B. METHODOLOGY

1) ARCHITECTURE PARADIGMS OF DL

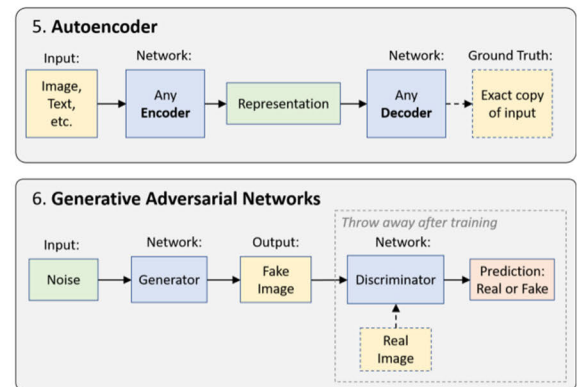
DL is referred to as a subfield of the ML domain based on ANN that emphasizes data representation learning [24]. The term “deep” in deep learning is understood as the depth of layers in a neural network [25]. Based on the representation learning, several typical DL models are briefly introduced in Fig. 3.

In this study, several architectural paradigms of DL that belong to the supervised learning class are of interest. The application of supervised learning to hydrological studies or streamflow forecasting could be explained by these studies have a tendency to require the input data to be hydrological

Supervised Learning



Unsupervised Learning



Reinforcement Learning

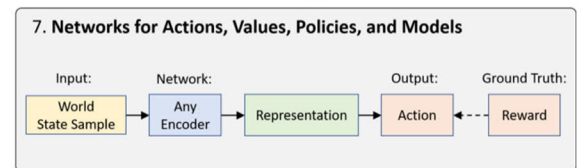


FIGURE 3. Overview of architectural paradigms of DL [26].

data such as water level, discharge, or precipitation [8]. Meanwhile, these data can only be obtained through monitoring and measuring by automatic or semi-automatic devices. For the streamflow-forecasting issue, six DL models—an FFNN model, a CNN model, and four RNN-based models—were built, and their performances were compared in this paper.

2) FFNN

FFNN, also known as multilayer perceptrons, is a form of ANN that consists of multiple layers of computing units, and data are processed in one direction (forward direction). For the FFNN model, the input variables in the input layer act as independent variables, and the output layer is referenced to the observed data. In addition to the input and output layers, a prevalent FFNN will consist of multiple hidden layers where the processing units in each of these hidden layers (neurons) are connected to all of the processing units in the previous hidden layer. There is no specific reference for choosing the number of hidden layers or the number of processing units in each hidden layer. Figure 4 shows a schematic of a typical FFNN.

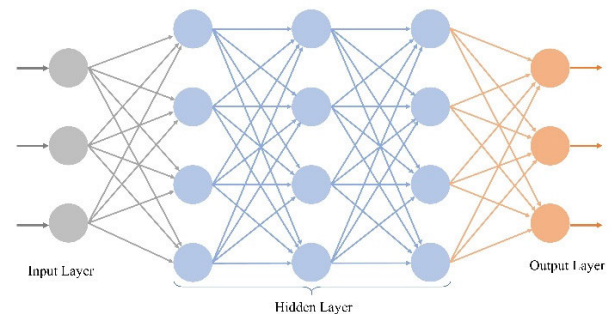


FIGURE 4. The structure of an FFNN with three hidden layers.

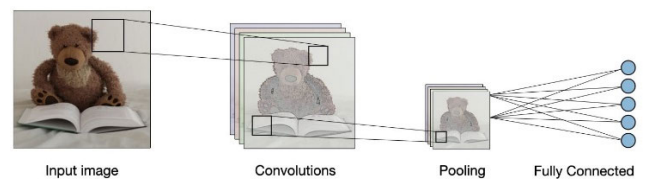


FIGURE 5. The structure of a traditional CNN [30].

3) CNN

CNN is a special FFNN that is designed with explicit assumptions toward working with two-dimensional input data structures, such as image data. CNNs have proven their effectiveness in handling dozens of critical tasks in the field of computer vision [27], [28]. Compared with conventional neural networks where hidden layers have densely connected computational units (neurons), CNN uses convolution layers to encode information, and computational units are arranged

in three dimensions, including width, height, and depth. Here, the depth is mentioned as the third dimension of an activation volume [29]. Fig. 5 depicts the typical architecture of a simple CNN.

There are three main types of layers in a CNN model: the convolution, pooling, and fully-connected layers. For a simple CNN model, the number of layers or the layer types might be flexibly stacked depending on the specific goal [31]. For more complex tasks, these layers can be used

together with other architectures, for example, LSTM [32] or autoencoder [31].

4) LSTM NEURAL NETWORK

The LSTM network architecture is considered an evolutionary form of the RNN architecture and was first introduced by Hochreiter and Schmidhuber [11]. Since then, LSTM has proven its effectiveness in solving complex tasks involving sequential data [7], [13], [33], [34]. A typical LSTM network consists of a repeating sequence of memory cells (blocks) arranged in a sequence referred to as the LSTM cells [35]. Fig. 6 shows the diagram of a memory cell in the LSTM network architecture.

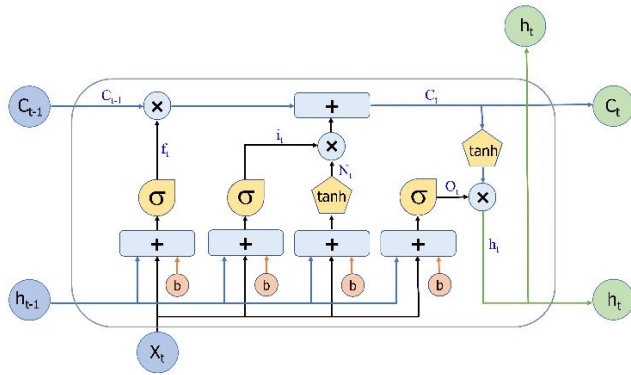


FIGURE 6. The typical structure of an LSTM cell [7].

Each memory cell in the LSTM network is interconnected through two vital components, where the hidden state (h_t) is known as short-term memory and the cell state (c_t) is referred to as long-term memory. By using the cell state, the LSTM network is able to store essential information for a prolonged period [7]. The cell state and hidden state values for an LSTM cell (at time step t) are represented by the following equations:

$$f_t = \sigma(U_f X_t + W_f h_{t-1}) \quad (1)$$

$$i_t = \sigma(U_i X_t + W_i h_{t-1}) \quad (2)$$

$$N_t = \tanh(U_n X_t + W_n h_{t-1}) \quad (3)$$

$$O_t = \sigma(U_o X_t + W_o h_{t-1}) \quad (4)$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes N_t \quad (5)$$

$$h_t = O_t \otimes \tanh(C_t) \quad (6)$$

Here, f_t , i_t , and O_t are values of forget gate, input gate, and output gate, respectively, at time step t ; N_t is the candidate of the cell state value; σ is a sigmoid activation function; U_j and W_j are weight matrices; and $\otimes$ denotes the element-wise multiplication.

5) GRU NEURAL NETWORK

First introduced in 2014 by Cho *et al.* [12], GRU is known as a simpler variant of LSTM because of their similarity in design and an equally excellent performance reported in several studies [36]. Instead of having three gating layers

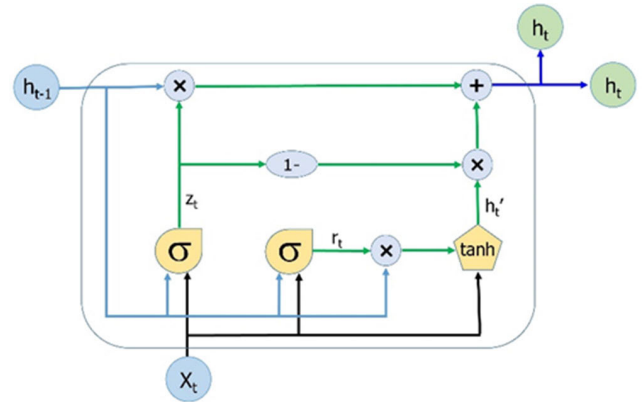


FIGURE 7. The typical structure of a GRU cell [14].

as in the LSTM architecture, its architecture has only two gating layers—the update gate (z_t) and the reset gate (r_t)—for processing information. Fig. 7 illustrates the architecture of a GRU memory cell.

For the GRU architecture, the output of the memory cell (h_t -hidden layer) at the t -th time step is used for both decision making and as input data for subsequent cells. The following equations determine this value:

$$z_t = \sigma(U_z X_t + W_z h_{t-1}) \quad (7)$$

$$r_t = \sigma(U_r X_t + W_r h_{t-1}) \quad (8)$$

$$h'_t = \tanh(U_h X_t + r_t \otimes W_h h_{t-1}) \quad (9)$$

$$h_t = (1 - z_t) \otimes h'_t + z_t \otimes h_{t-1} \quad (10)$$

where h'_t is the candidate of the hidden state at time step t ; U_j and W_j are weight matrices; and $\otimes$ denotes the element-wise multiplication.

6) STACKED LSTM NEURAL NETWORK

Contrary to the original LSTM model with only a single hidden layer, the StackedLSTM network model is an enhancement of the LSTM architecture with multiple hidden layers, where each hidden layer consists of multiple memory cells. The application of StackedLSTM has been demonstrated in several complex tasks such as machine translation [37] or speech recognition [34]. However, the exploitation of deep LSTM architecture for problems in the hydrological field is modest compared with other disciplines. According to Pascanu *et al.* [38], the stacking of multiple hidden layers will make the model deeper (deep LSTM) and allow the hidden state at each level to operate more efficiently at various timescales. Aim at examining the necessity of using deep LSTM models (multiple hidden layers) in streamflow forecasting, a deep LSTM network architecture with stacking of three hidden layers is proposed in this study. Fig. 8 shows an illustrative example of the StackedLSTM model.

7) BILSTM NEURAL NETWORK

For conventional LSTM (or standard RNN) models, data processing is performed in one direction (forward) and essential

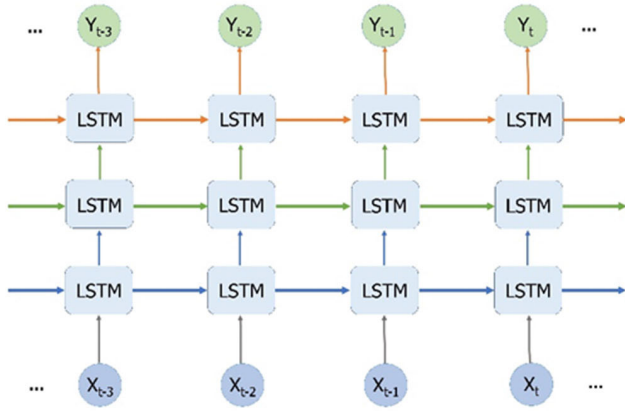


FIGURE 8. The structure of a StackedLSTM neural network with three hidden layers.

information is stored (or forecasted) based on previous data. This is one of the shortcomings of the standard LSTM network because it is not possible to obtain future input information from the current state. In other words, the output layer information can only be reached by exploiting previous contexts [34]. BiLSTM is an advanced extension of the LSTM architecture, which processes data simultaneously in both forward and backward directions. The output of the BiLSTM model at time step t is generated based on a combination of two separate hidden layers, where each hidden layer represents a training sequence, corresponding to the forward and backward directions [34]. Therefore, BiLSTM is expected to capture both the future and past information of the input sequence better than the standard LSTM [39]. Fig. 9 illustrates a typical structure for a BiLSTM model.

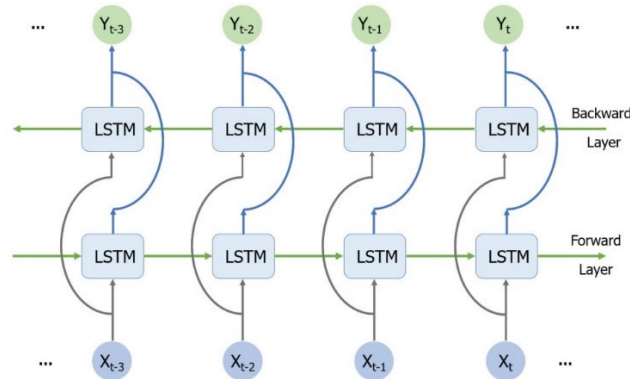


FIGURE 9. The structure of a BiLSTM neural network.

C. PERFORMANCE METRIC INDEX

In this study, three statistical indicators, Nash–Sutcliffe efficiency (NSE), mean absolute error (MAE), and root mean square error (RMSE), were used to assess the predictive performance of the DL models. These are prevalent metrics widely used in statistical and hydrological models to evaluate the correlation between two data series.

According to Nash and Sutcliffe [40], the NSE value measures the difference between variables and the mean and gives the ratio of the initial variance calculated from the model. The NSE value ranges from $-\infty$ to 1. The MAE function computes the mean of the absolute error over the test sample between forecasted and observed values. This value varies between 0 and ∞ , and lower values are better. In addition, the RMSE parameter is often used to evaluate how the forecasted data match the observed data based on the relative range of the data, and an RMSE equal to zero implies a perfect fit. While all the individual differences in the MAE are weighted equally, the RMSE value tends to give a relatively high weight for the individual differences that have a large error. These indicators are calculated as follows:

$$NSE = 1 - \frac{\sum_{i=1}^n (x_i - y_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - y_i| \quad (12)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \quad (13)$$

In the above equations, x_i and y_i are observed and forecasted discharges at time t , respectively; $\bar{x}$ is the mean of observed discharges; and n is the total number of observations.

III. MODEL STRUCTURE

To comprehensively evaluate the performance of models in the supervised learning class for the streamflow-forecasting problem, six DL models were developed and objectively assessed in this study based on the programming language—Python [41]—and several open-source software libraries such as TensorFlow [42] and Keras [43]. However, the parameters within each of these models, depending on the network architecture, may have slight differences from the other models, especially in the FFNN and CNN models.

In the case of the FFNN model, a neural network model with three hidden layers stacked in order has been proposed. In addition, the recommended number of computation units for each hidden layer ranges from 128 to 2,048 (in terms of a power function of 2). The aim of selecting parameter values in the neural network model as a power function of 2 is towards saving the computational resources of these models in the processing of input data. Besides, there is a minor adjustment to the input data format of the FFNN model to help the model effectively learn dependencies in the sequential data problem. At each given time step, the raw data are formatted into the observed data series of two timesteps (observed data at time steps t and $t-1$) instead of just using data are observed at time step t .

For the CNN model, the recommended architecture in this study is an ordered combination of two convolutional

layers (Conv1D), one pooling layer (MaxPooling1D), and one fully-connected layer. To simplify the construction of model parameters, the number of filters in each convolution layer and the number of units in the fully-connected layer are selected as the same, and this value ranges from 8 to 128 in terms of powers of 2. For the pooling layer, the MaxPooling1D function with a pool size value of 2 (pool_size=2) was used to halve the dimension of feature maps that were generated after the convolutional operation. In addition, a small modification of the raw data format was made to enhance the efficiency of the CNN model when working with sequential data. Similar to the FFNN model, the input data size at any given time step (t) of the CNN model has been formatted as a series of the eight most recent observed values.

The remaining four DL models—LSTM, GRU, StackedLSTM, and BiLSTM—have certain architectural similarities because they are designed to overcome the shortcomings of the RNN model. Similar to the CNN model, the available observational data are formatted into data series before they are fed as input to these models for the training process. Accordingly, each data series represents data of the last eight observed times (from $t-7$ to t). In addition, the recommended number of units per hidden layer is a power function of 2 and ranges from 8 to 128. In contrast to the LSTM and GRU models, which use just one hidden layer in their structure, the StackedLSTM model has more complexity with the successive stacking of three hidden layers. Moreover, the BiLSTM model is an enhanced LSTM model and is more complicated than the StackedLSTM model. Although its architecture consists of just two hidden layers compared to the three hidden layers of the StackedLSTM, its hidden layers operate separately from each other, and they represent two directions of data training—forward and backward (Fig. 9).

In addition to the setting of internal parameters of the models, training parameters (hyperparameters) such as optimization algorithm, loss function, and learning rate are vital components of DL models. The Adam optimization algorithm [44] with a default-learning rate of 0.001 is used throughout this study. This is also one of the algorithms widely applied in DL studies [45]. Along with the Adam optimization algorithm, the mean squared error function is the recommended loss function in the six DL models because of its effectiveness in hydrologic studies [7], [31].

Besides, for the DL models to capture the essential information during the training phase, the recommended number of epochs in the six models is 10,000 (this is the maximum value that can be reached if the model does not converge). Several other advanced techniques are applied in this study to improve the efficiency of the DL models. For instance, early stopping and dropout techniques [46] to prevent overfitting problems (if possible) and model checkpoint techniques to record critical information of the model during the training stage. Further, the early stopping technique is employed to stop the training process of these models if the values of loss functions on the validation dataset do not decrease after 2,000 consecutive epochs.

IV. RESULTS AND DISCUSSION

DL models have been carefully constructed and evaluated through three processes of training, validating, and testing, respectively. The division of the sizes of the datasets has been discussed in the above sections. In this section, this study only presents the results of the quantitative comparison between the observed and forecasted value series corresponding to the validation and testing phases. Accordingly, the rest of the first dataset (accounting for 20% of 19 years) was used to validate the performance of DL models in predicting flowrate one day and two days in advance. This is also a data set that has not been used in the training phase. For the testing phase, the data series of 2014 (previously unseen dataset) was exploited to objectively verify the performance of DL models.

A. VALIDATION RESULTS

From the recommended parameters and model structures, the six DL models have undergone training and validation to provide one- and two-day advance forecasts. During the validation phase, the performance of each DL model is assessed through the statistical indicators MAE, RMSE, and NSE, respectively. The one- and two-day flowrate forecast results at the Son Tay hydrological station of six models are demonstrated in Tables 2 and 3.

From the figures in Table 2, the DL models exhibit impressive performance even when the inner parameters of the model are changed. Here, the unit number is a term that corresponds to a parameter of the LSTM, GRU, StackedLSTM, and BiLSTM models, while the number of neurons is a parameter of the CNN and FFNN models. Models in the RNN family (including LSTM, GRU, StackedLSTM, and BiLSTM) demonstrate stability, where the difference in the performance of the forecast cases is insignificant, and the NSE value in most of these cases is above 0.98. Particularly, the LSTM model can produce better forecast results, corresponding to an NSE of 0.988 (case of LSTM 16) and an MAE value of 150 m³/s, followed by StackedLSTM and BiLSTM models with an NSE value of 0.986 and an MAE value of approximately 160 m³/s. The remaining two models (CNN and FFNN) exhibit lower performance and uncertainty in the forecasted cases than models in the RNN family. Although the best forecast results of these two models are quite competitive, as the NSE value is 0.978 and 0.982 for CNN and FFNN, respectively. Table 2 also shows the fluctuation in the model performance when the number of neurons is changed.

For the two-day flowrate forecast ahead (Table 3), a similar trend was recorded as the six models illustrated their competitive performance. The LSTM and GRU models outperformed the rest, with the NSE coefficient, which shows consistency with the observed and predicted data from the two models up to 0.93 (in the case of LSTM 16 and GRU 8). Meanwhile, the StackedLSTM and BiLSTM models continue to exhibit impressive stability and performance, with NSE reaching 0.92 (in StackedLSTM 8 and BiLSTM 8), which is slightly

TABLE 2. The parameters and performance of six models for one-day advance forecast (validation period).

Model	No.	Case	Number of neurons/units	Number of epochs	MAE (m ³ /s)	RMSE (m ³ /s)	NSE
FFNN	1	FFNN 128	128	522	326.6	469.5	0.956
	2	FFNN 256	256	633	313.1	403.9	0.967
	3	FFNN 512	512	421	185.6	320.6	0.979
	4	FFNN 1024	1024	508	211.2	296.8	0.982
	5	FFNN 2048	2048	35	243.4	362.3	0.974
CNN1D	6	CNN 8	8	1,913	303.6	406.8	0.967
	7	CNN 16	16	1,080	229.8	355.0	0.975
	8	CNN 32	32	538	211.1	332.8	0.978
	9	CNN 64	64	215	218.8	365.5	0.973
	10	CNN 128	128	65	237.3	408.7	0.967
LSTM	11	LSTM 8	8	682	210.6	313.8	0.980
	12	LSTM 16	16	544	149.9	248.0	0.988
	13	LSTM 32	32	478	165.2	277.6	0.985
	14	LSTM 64	64	200	194.5	292.5	0.983
	15	LSTM 128	128	206	204.8	349.0	0.976
GRU	16	GRU 8	8	670	162.2	282.9	0.984
	17	GRU 16	16	602	200.1	301.0	0.982
	18	GRU 32	32	403	174.5	304.5	0.981
	19	GRU 64	64	241	197.6	347.6	0.976
	20	GRU 128	128	226	178.4	295.1	0.983
Stacked LSTM	21	Stacked 8	8	561	190.7	319.8	0.980
	22	Stacked 16	16	458	180.9	295.6	0.983
	23	Stacked 32	32	433	159.4	260.6	0.986
	24	Stacked 64	64	464	169.6	295.5	0.983
	25	Stacked 128	128	122	179.7	302.0	0.982
BiLSTM	26	BiLSTM 8	8	487	162.8	268.9	0.986
	27	BiLSTM 16	16	213	191.1	321.2	0.979
	28	BiLSTM 32	32	255	215.5	370.2	0.973
	29	BiLSTM 64	64	96	205.8	312.8	0.980
	30	BiLSTM 128	128	107	206.1	326.1	0.979

lower than those of the LSTM and GRU models but higher than those of the CNN and FFNN models (NSE is only about 0.90).

In addition, the figures in Tables 2 and 3 reveal that the complexity of the models (when increasing the number of units/neurons) does not improve model performance. Theoretically, increasing the number of units per cell of the LSTM,

GRU, StackLSTM, and BiLSTM models or increasing the number of neurons in the CNN and FFNN models will make these models more complicated due to a large number of parameters to be estimated. However, it seems that DL models, especially those belonging to the RNN family, tend to be more efficient when the number of units/neurons in the range varies from 8 to 32 (except for the FFNN model). For the

TABLE 3. The parameters and performance of six for two-day advance forecast (validation period).

Model	No.	Case	Number of neurons/units	Number of epochs	MAE (m ³ /s)	RMSE (m ³ /s)	NSE
FFNN	31	FFNN 128	128	35	506.0	742.2	0.890
	32	FFNN 256	256	79	455.9	677.0	0.908
	33	FFNN 512	512	35	404.3	676.1	0.908
	34	FFNN 1024	1024	64	448.1	770.1	0.881
	35	FFNN 2048	2048	35	426.4	692.4	0.904
CNN1D	36	CNN 8	8	779	484.8	719.1	0.897
	37	CNN 16	16	246	434.4	729.2	0.894
	38	CNN 32	32	50	447.3	774.4	0.880
	39	CNN 64	64	38	499.5	831.5	0.862
	40	CNN 128	128	30	480.4	751.3	0.887
LSTM	41	LSTM 8	8	187	392.3	620.9	0.923
	42	LSTM 16	16	291	355.8	591.4	0.930
	43	LSTM 32	32	143	424.1	721.7	0.896
	44	LSTM 64	64	53	440.5	710.3	0.899
	45	LSTM 128	128	38	480.4	742.4	0.890
GRU	46	GRU 8	8	1,482	354.1	600.6	0.928
	47	GRU 16	16	112	365.8	630.1	0.921
	48	GRU 32	32	80	416.7	684.5	0.907
	49	GRU 64	64	66	484.9	796.7	0.873
	50	GRU 128	128	9	442.9	680.9	0.907
Stacked LSTM	51	Stacked 8	8	454	380.1	621.3	0.923
	52	Stacked 16	16	207	396.8	648.4	0.916
	53	Stacked 32	32	77	419.1	684.6	0.906
	54	Stacked 64	64	50	457.5	734.0	0.892
	55	Stacked 128	128	37	470.4	723.7	0.895
BiLSTM	56	BiLSTM 8	8	160	378.4	637.8	0.919
	57	BiLSTM 16	16	160	439.6	750.2	0.888
	58	BiLSTM 32	32	100	559.7	801.4	0.872
	59	BiLSTM 64	64	43	476.2	728.9	0.894
	60	BiLSTM 128	128	77	444.9	687.5	0.906

FFNN model, the optimal number of neurons in this study is 1,024 and 512, corresponding to the one- and two-day prediction cases, respectively.

The validation step results have described the effectiveness of the six models in predicting runoff one day and two days in advance at the Son Tay hydrological station. This process is obtained by applying advanced techniques, such as early

stopping and model checkpoint, to the models. The application of these techniques will help save computer resources by stopping the active training process, and thus, DL models will operate more stably and efficiently. Accordingly, the six proposed models are set to stop the training process if the error of the loss function did not improve after 2,000 calculations and updates of weight continuously (the number of epochs).

TABLE 4. Summary of performance of six models for two forecast cases in the testing step.

Forecast for	No.	Case	Number of neurons/units	Number of epochs	MAE (m ³ /s)	RMSE (m ³ /s)	NSE
One day	4	FFNN 1024	1024	508	187.2	230.4	0.969
	8	CNN 32	32	538	195.7	267.1	0.957
	12	LSTM 16	16	544	119.4	160.6	0.985
	16	GRU 8	8	670	103.6	141.9	0.988
	23	Stacked 32	32	433	140.5	205.3	0.975
	26	BiLSTM 8	8	487	129.2	177.9	0.981
Two days	33	FFNN 512	512	35	375.8	519.5	0.840
	36	CNN 8	8	779	421.7	719.1	0.897
	42	LSTM 16	16	291	269.2	591.4	0.930
	46	GRU 8	8	1,482	252.6	600.6	0.928
	51	Stacked 8	8	454	293.4	621.3	0.923
	56	BiLSTM 8	8	160	317.6	637.8	0.919

This explains why the number of epochs of each forecasting case is different (see Tables 2–3). Besides, another trend observed is the inverse relationship between the number of epochs and the number of units/neurons in Tables 3–4. This means that the DL models tend to reach their optimal value faster as the model complexity increases or the convergence rate of the models is proportional to their complexity.

B. TESTING RESULTS

After training and validation processes, the best parameter set of the model corresponding to the forecasted cases of the six models was selected. The remaining dataset of 2014, an independent dataset, was used as an unseen dataset previously to objectively verify the performance of these models with two forecast cases, for the one- and two-day ahead at Son Tay gauge station. Table 4 summarizes the results, and Figures 10–12 show the visual comparisons between the observed and forecasted data of these models.

From the results described during the testing period in Table 4, the six models still show their effectiveness in the two cases of discharge forecast for Son Tay station. The RNN family models generally exhibit better performance than the other two models (FFNN and CNN). As such, the LSTM and GRU are two models that produce equally excellent performance when NSE coefficients are about 0.988 and 0.930, corresponding to one- and two-day discharge forecasts, respectively (case of LSTM 16 and GRU 8). In addition to the LSTM and GRU models, the statistical indicators recorded notable results for the StackedLSTM and BiLSTM models, where the NSE values corresponding to the two forecast cases were nearly 0.98 and 0.92, respectively. For the FFNN and CNN models, although noticeable results were observed during the validation step, it seems that the performance of these models

decreased slightly during the testing step. In the FFNN model, the NSE values in the testing period corresponding to the two forecast cases are only about 0.97 and 0.84, respectively, compared to the values of 0.98 and 0.90 during the validation period.

For the one-day flowrate forecast, the correlation between observed data and forecasted data from the six DL models is indicated by the scatter plot in Fig. 10 and the violin plot in Fig. 11. In Fig. 10, each pair of data represented by a dot on the chart, and a model is considered effective if the dots are close to the 45-degree diagonal. The LSTM and GRU models are more prominent than the other four when the forecasted discharge is close to the observed discharge. This is noted more clearly in flood season runoff projections, where the observed runoff values tend to be large. Besides, another statistical comparison is visualized by the violin plot (Fig. 11) to evaluate the performance of DL models when compared with observed values. In addition to illustrating statistical indicators such as quartile ranges or mean values of predictive data series from DL models, the violin plot provides information about the shape and probability density distribution of these data series. The depicted results in the violin diagrams indicated consistency with the figure-based analysis in Table 4 and the scatter plot in Fig. 10. Accordingly, the LSTM and GRU models recorded more stable and better performance than the rest.

For LSTM and GRU models, Fig. 12 is obvious evidence that it is difficult to quantify the differences between these two models due to a close consistency with the observed data for the one-day forecast ahead. Both models could produce forecasts with competing performance. Further, although the correlation coefficient (NSE value) of these two models (compared to the observed data) was close to

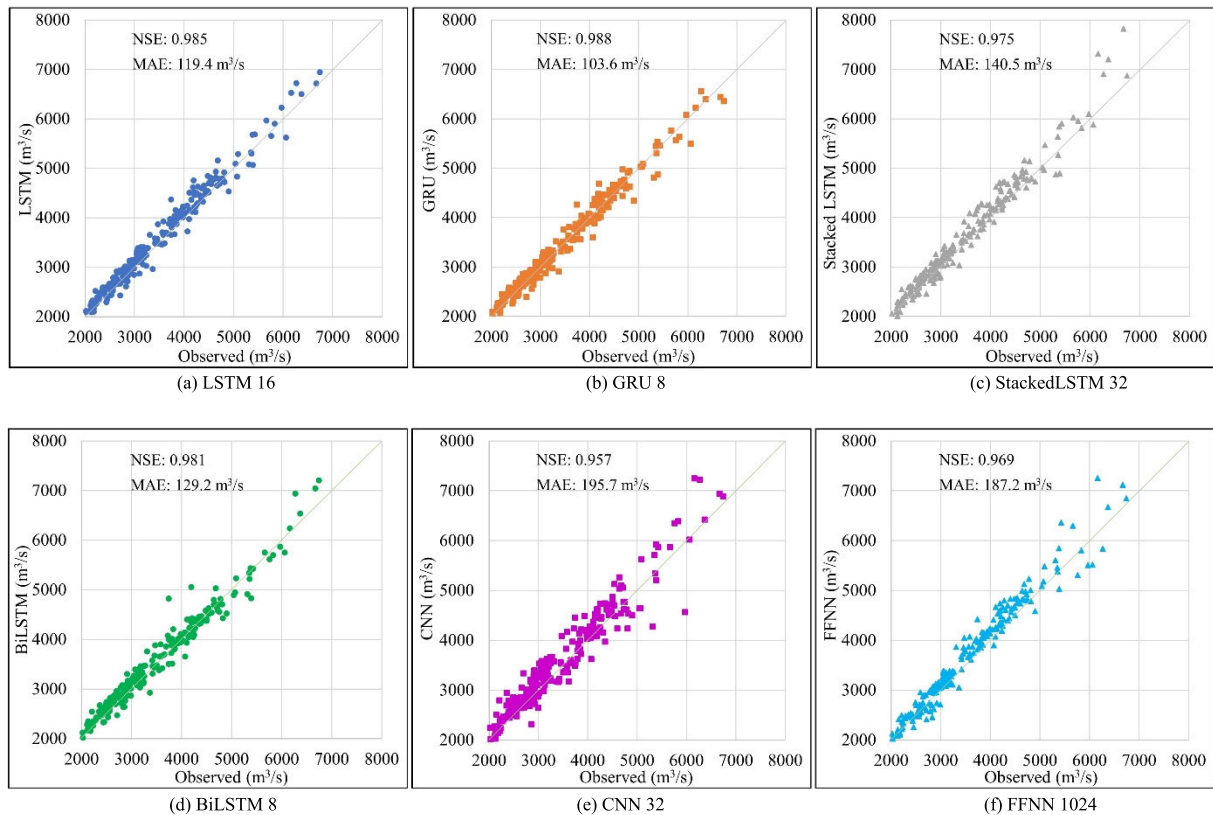


FIGURE 10. Scatter plot of one-day flowrate forecasting in the testing step for: (a) LSTM 16, (b) GRU 8, (c) StackedLSTM 32, (d) BiLSTM 8, (e) CNN 32, (f) FFNN 1024.

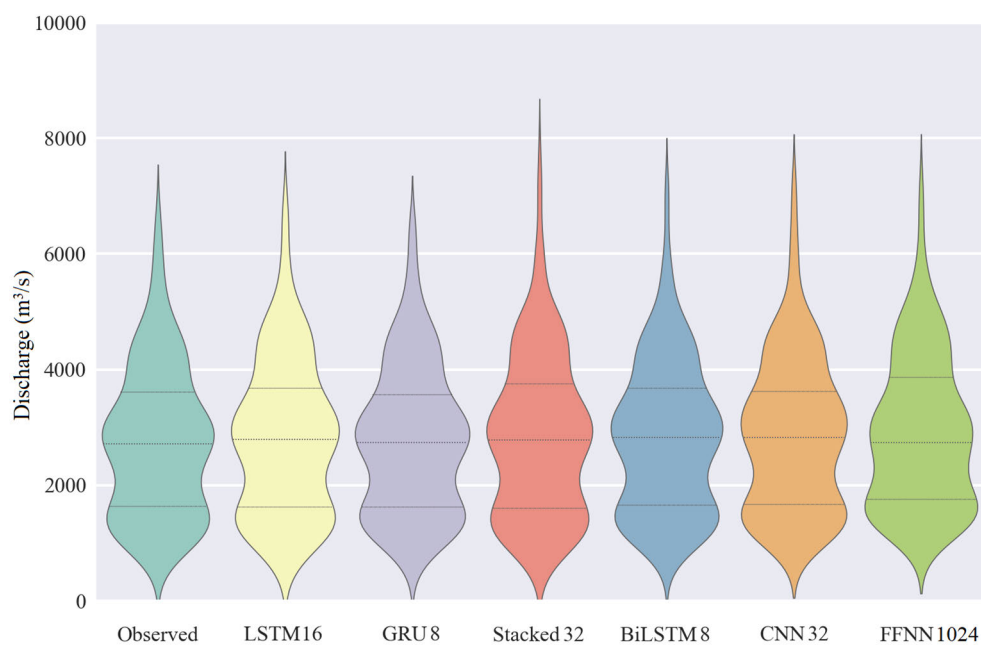


FIGURE 11. Violin plot of one-day flowrate forecasting in the testing step.

0.988 in both evaluation periods in the validation phase, the GRU model slightly outperformed the LSTM model in the testing phase.

Generally, both LSTM and GRU excellently demonstrate their streamflow predictive ability as well as capture flood processes. Table 5 depicts a more specific comparison

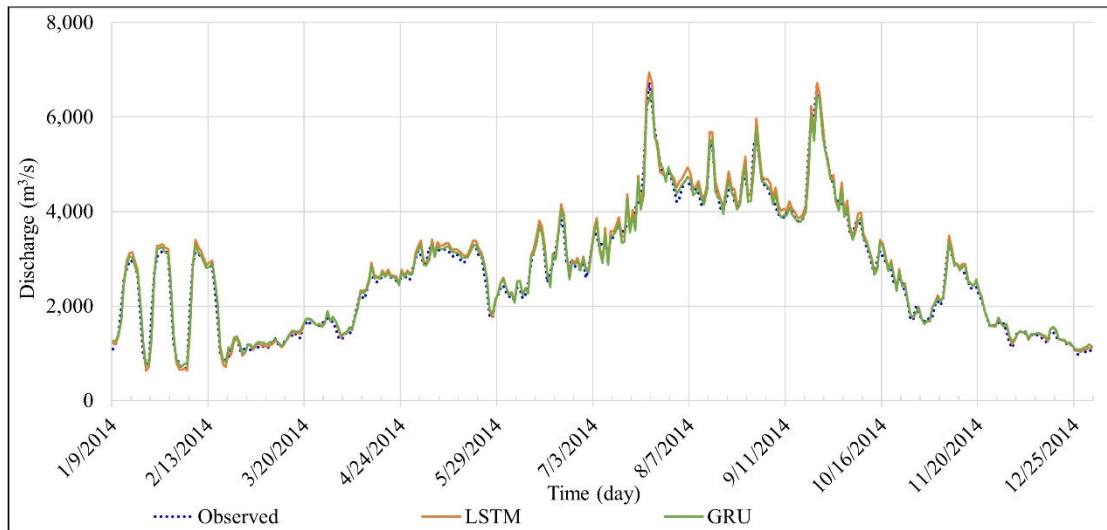


FIGURE 12. Comparison of the observed and forecasted values for one-day forecasting (cases LSTM 16 and GRU 8).

TABLE 5. Results of flood peak forecasting for one-day in the testing step on July 23.

Case	Forecasted Peak	Observed Peak	Absolute error (m^3/s)	Relative error (%)
LSTM 16	6945.1	6740	205.1	3.0%
GRU 8	6355.4	6740	-384.6	-5.7%
Stacked 32	6870.7	6740	130.7	1.9%
BiLSTM 8	7204.5	6740	464.5	6.9%
CNN 32	6886.0	6740	146.0	2.2%
FFNN 1024	6847.1	6740	107.1	1.6%

of 2014 when the flood forecasting capability of the proposed models is of interest. This is also the year of the complicated flood process because the observed flowrate values at Son Tay station recorded two flood peaks occurring on July 23 and September 22 (higher values on July 23).

For the forecasted flood peak on July 23, most of the DL models illustrated their advantages in flood predictive ability, and the flood peaks are forecasted to appear simultaneously. The relative errors identified for these models in the one-day forecast ahead ranged from 1.6% to 5.7% compared with the observation value recorded at the Son Tay station of $6,740 \text{ m}^3/\text{s}$. For the LSTM and GRU models, these figures are 3.0% and 5.7%, respectively, with absolute errors of about 205 and $385 \text{ m}^3/\text{s}$, respectively. This is an acceptable tolerance in the event of a flood. These results prove that the six models could learn data dependence effectively.

Besides, several various scenarios of data have been employed to comprehensively evaluate the performance of the six models. For instance, the sizes of the training and

validation datasets were further redistributed in a manner that decreases the size of the training dataset and increases the size of the validation dataset. This aimed to evaluate the size and quality of the data for the training phase. This study's results reveal that the simple models are more stable and efficient with current data division than others. Moreover, the difference (if possible) is minute, with the accuracy fluctuating by approximately 1%–2% (depending on the forecasted cases) from that selected above.

V. CONCLUSION

In this study, six DL models were developed to forecast streamflow one and two days in advance at the Son Tay hydrological station on the Red River. Besides, the performance of these models is carefully investigated through training, validation, and testing processes. The input data of these models for forecasting at the Son Tay gauge station are a series of observed flowrate data at seven hydrological stations on three major river branches of the Red River system—Thao River, Da River, and Lo River. Other data on precipitation, water levels, topographic features, and land use are not required in this study.

For a comprehensive comparison of the performance of DL models (belonging to the supervised learning class), the inner parameters of models have been set up as the same. These parameters include the optimization algorithm, learning rate, loss function, activation function, the number of units per memory cell (for models in the RNN family), or number of neurons (for FFNN and CNN models). The validation and testing step results indicate that all of them perform well in one-day and two-day discharge forecasts in advance. The model performances will decrease as the required forecast period increases. Of which, four models in the RNN family exhibited considerably better performance and maintained stability under better predictive cases than the CNN and

FFNN models, especially in the case of a two-day forecast (or a multi-day forecast ahead). This can also be explained because the model architectures that belong to the RNN class (e.g., LSTM, GRU, or BiLSTM) are essentially designed to address problems involving sequential data effectively. Meanwhile, CNN architecture is often widely employed in the field of computer vision, and FFNN is a form of deep neural network that can be used in various fields.

Another conclusion that can be drawn from the RNN family models is that their complexity is not accompanied by performance improvement. An illustrative example of this would be the case of the StackedLSTM and BiLSTM models. These are models with significantly high complexity, but their respective performance is not higher than the two simple models—LSTM and GRU. Besides, the model construction and training process observed that the two complex models tend to require a substantial amount of computational resources compared with LSTM and GRU (computation time required per epoch increases significantly). In the context of hydrological forecasting problems, simple architectural models such as LSTM and GRU (with only one hidden layer) are sufficient to produce highly reliable forecasts while minimizing computation time. In addition to the structural complexity of the models, the number of units per memory cell is a vital quantity in the RNN-based models due to their direct influence on the data processing performance of the model and computation time per epoch. The empirical reviews in this study indicate that the recommended number of units in each memory cell range of 8–32 is consistent with hydrological prediction problems. The parameter increase does not seem to improve the performance of the models (see Tables 3 and 4), which may even lead to overfitting problems in certain cases.

For the LSTM and GRU models, an equally excellent performance of the two models in streamflow and flood peak forecasting was demonstrated and mentioned in the discussion section. This can be explained because the GRU architecture is known as a simpler variant of the LSTM architecture, and the LSTM model tends to represent slightly better performance than a GRU when working with vast input data. When the model architecture is simple, or the number of units per memory cell is small, the LSTM model indicates the computation speed per epoch slightly faster than the GRU model. By contrast, the GRU model exhibits a slightly faster computation speed if the model is more complexly established with more units in each memory cell. LSTM has been employed in various domains and receives a myriad of support from various programming languages and frameworks. Moreover, these two models are designed similarly, thus, the choice of model to exploit will depend on the purpose of the user.

Another key factor that has a major influence on the effectiveness of DL models is the characterization of the input data. According to Le et al. [7], the kind of data has a more substantial impact on performance than the data quantity, and models tend to operate more efficiently when

the correlation coefficient between the data at the observation stations and the target station is higher. This is why this study only focuses on exploiting the discharge data without using them along with other data such as water level or precipitation data. Moreover, it is sometimes challenging to collect various data types simultaneously and at all hydrological stations, especially in real-time streamflow forecasting.

The data used in this study were daily observed flowrate data at seven hydrological stations, and DL models have been established to provide streamflow forecasts for one-day and two-day periods in advance. Besides, these models can be built to produce forecasts with hourly scales (for example, one hour or two hours ahead) if the collected input data are hourly observational data. For hourly scale data or observed hourly data, empirical results indicate that RNN-based models exhibit superiority over the CNN and FFNN models, especially in those cases where predicting multiple time steps ahead is of interest. Here, each time step is understood as one-time observed data is recorded.

DL models, generally, belong to the data-driven model type, which can produce forecasts with considerable accuracy at one or several fixed sites without requiring various input data types compared with conventional hydrodynamic models. Although these models have demonstrated their superiority in providing highly efficient short-term streamflow predictions, the role of hydrological models in the physical simulation of rainfall–runoff processes cannot be ignored. Therefore, hybrid models that combine DL and hydrological models should be exploited more to solve the problem of long-term streamflow forecast more effectively.

To our best knowledge, this study is the first step toward enhancing the understanding of exploiting DL techniques in the streamflow-forecasting problem. The findings of this study have pointed out the advantages, shortcomings, and specific appraisal of the performance of the six DL models in the short-term hydrological prediction problem. As a result, this study can be a valuable reference for research on streamflow forecasting on the Red River system in Vietnam and other major river basins globally.

REFERENCES

- [1] CRED. (2019). *Natural Disasters 2018*. Centre for Research on the Epidemiology of Disasters CRED, Brussels, Belgium. [Online]. Available: <https://www.cred.be/sites/default/files/CREDNaturalDisaster2018.pdf>
- [2] CRED. (2020). *Natural Disasters 2019: Now is the Time to Not Give Up*. Centre for Research on the Epidemiology of Disasters—CRED, Brussels, Belgium. [Online]. Available: https://emdat.be/sites/default/files/adsr_2019.pdf
- [3] J. Marçais and J.-R. de Dreuzy, “Prospective interest of deep learning for hydrological inference,” *Groundwater*, vol. 55, no. 5, pp. 688–692, Sep. 2017, doi: [10.1111/gwat.12557](https://doi.org/10.1111/gwat.12557).
- [4] S. Ardabili, A. Mosavi, M. Dehghani, and A. R. Várkonyi-Kóczy, “Deep learning and machine learning in hydrological processes climate change and earth systems a systematic review,” in *Engineering for Sustainable Future*, Cham, Switzerland: Springer, 2019, pp. 52–62, doi: [10.1007/978-3-030-36841-8_5](https://doi.org/10.1007/978-3-030-36841-8_5).
- [5] A. Mosavi, P. Ozturk, and K.-W. Chau, “Flood prediction using machine learning models: Literature review,” *Water*, vol. 10, no. 11, p. 1536, Oct. 2018, doi: [10.3390/w10111536](https://doi.org/10.3390/w10111536).

- [6] G. Lee, "Assessment of prediction uncertainty due to various sources involved in rainfall-runoff modeling," Ph.D. dissertation, Disaster Prevention Res. Inst., Kyoto Univ., Kyoto, Japan, 2008.
- [7] X. H. Le, H. V. Ho, G. Lee, and S. Jung, "Application of long short-term memory (LSTM) neural network for flood forecasting," *Water*, vol. 11, no. 7, p. 1387, Jul. 2019, doi: [10.3390/w11071387](https://doi.org/10.3390/w11071387).
- [8] X. H. Le, "Application of deep neural network for spatiotemporal hydrological data analysis," Ph.D. dissertation, Dept. Disaster Prevention Environ. Eng., Kyungpook Nat. Univ., Daegu, South Korea, 2020.
- [9] Y. Bengio, P. Simard, and P. Frasconi, "Learning long-term dependencies with gradient descent is difficult," *IEEE Trans. Neural Netw.*, vol. 5, no. 2, pp. 157–166, Mar. 1994, doi: [10.1109/72.279181](https://doi.org/10.1109/72.279181).
- [10] F. K. John and C. K. Stefan, "Gradient flow in recurrent nets: The difficulty of learning long-term dependencies," in *A Field Guide to Dynamical Recurrent Networks*. Piscataway, NJ, USA: IEEE, 2001.
- [11] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735).
- [12] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," 2014, *arXiv:1406.1078*. [Online]. Available: <http://arxiv.org/abs/1406.1078>
- [13] F. Kratzert, D. Klotz, C. Brenner, K. Schulz, and M. Hermegger, "Rainfall-runoff modelling using long short-term memory (LSTM) networks," *Hydrol. Earth Syst. Sci.*, vol. 22, no. 11, pp. 6005–6022, Nov. 2018, doi: [10.5194/hess-22-6005-2018](https://doi.org/10.5194/hess-22-6005-2018).
- [14] X. H. Le, H. V. Ho, and G. Lee, "Application of gated recurrent unit (GRU) network for forecasting river water levels affected by tides," in *Proc. APAC*. Singapore: Springer, 2020, pp. 673–680, doi: [10.1007/978-981-15-0291-0_92](https://doi.org/10.1007/978-981-15-0291-0_92).
- [15] X. H. Le, S. Jung, M. Yeon, and G. Lee, "River water level prediction based on deep learning: Case study on the Geum River, South Korea," in *Proc. ICSCSE*, in Lecture Notes in Civil Engineering, vol. 145. Singapore: Springer, 2021, pp. 319–325, doi: [10.1007/978-981-15-0291-0_92](https://doi.org/10.1007/978-981-15-0291-0_92).
- [16] X. H. Le, "Reconstruction of missing daily streamflow data using recurrent neural network," (in Vietnamese), *J. Water Resour. Environ. Eng.*, vol. 66, pp. 63–70, 2019.
- [17] F. Kratzert, D. Klotz, M. Hermegger, A. K. Sampson, S. Hochreiter, and G. S. Nearing, "Toward improved predictions in ungauged basins: Exploiting the power of machine learning," *Water Resour. Res.*, vol. 55, no. 12, pp. 11344–11354, Dec. 2019, doi: [10.1029/2019WR026065](https://doi.org/10.1029/2019WR026065).
- [18] S. Miao and W.-H. Hung, "River flooding forecasting and anomaly detection based on deep learning," *IEEE Access*, vol. 8, pp. 198384–198402, 2020, doi: [10.1109/ACCESS.2020.3034875](https://doi.org/10.1109/ACCESS.2020.3034875).
- [19] D. Liu, W. Jiang, L. Mu, and S. Wang, "Streamflow prediction using deep learning neural network: Case study of yangtze river," *IEEE Access*, vol. 8, pp. 90069–90086, 2020, doi: [10.1109/ACCESS.2020.2993874](https://doi.org/10.1109/ACCESS.2020.2993874).
- [20] L. Ni, D. Wang, V. P. Singh, J. Wu, Y. Wang, Y. Tao, and J. Zhang, "Streamflow and rainfall forecasting by two long short-term memory-based models," *J. Hydrol.*, vol. 583, Apr. 2020, Art. no. 124296, doi: [10.1016/j.jhydrol.2019.124296](https://doi.org/10.1016/j.jhydrol.2019.124296).
- [21] X. Yuan, C. Chen, X. Lei, Y. Yuan, and R. M. Adnan, "Monthly runoff forecasting based on LSTM-ALO model," *Stochastic Environ. Res. Risk Assessment*, vol. 32, no. 8, pp. 2199–2212, Aug. 2018, doi: [10.1007/s00477-018-1560-y](https://doi.org/10.1007/s00477-018-1560-y).
- [22] F. Kratzert, D. Klotz, G. Shalev, G. Klambauer, S. Hochreiter, and G. Nearing, "Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets," *Hydrol. Earth Syst. Sci.*, vol. 23, no. 12, pp. 5089–5110, Dec. 2019, doi: [10.5194/hess-23-5089-2019](https://doi.org/10.5194/hess-23-5089-2019).
- [23] X. H. Le, H. V. Ho, and G. Lee, "River streamflow prediction using a deep neural network: A case study on the Red River, Vietnam," *Korean J. Agric. Sci.*, vol. 46, no. 4, pp. 843–856, 2019, doi: [10.7744/kjoas.20190068](https://doi.org/10.7744/kjoas.20190068).
- [24] F. Chollet, *Deep Learning With Python*. New York, NY, USA: Manning Publications, 2017.
- [25] E. Kavakoglu, *AI vs. Machine Learning vs. Deep Learning vs. Neural Networks: What's the Difference?*. IBM, Armonk, NY, USA. Accessed: Jan. 10, 2021. [Online]. Available: <https://www.ibm.com/cloud/blog/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks>
- [26] L. Fridman, *MIT Deep Learning Basics: Introduction and Overview with TensorFlow*. Accessed: Oct. 10, 2019. [Online]. Available: <https://medium.com/tensorflow/mit-deep-learning-basics-introduction-and-overview-with-tensorflow-355bcd26baf0>
- [27] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015, doi: [10.1038/nature14539](https://doi.org/10.1038/nature14539).
- [28] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, "Object detection with deep learning: A review," 2018, *arXiv:1807.05511*. [Online]. Available: <http://arxiv.org/abs/1807.05511>
- [29] A. Karpathy, *CS231n: Convolutional Neural Networks for Visual Recognition*. Accessed: Sep. 10, 2019. [Online]. Available: <http://cs231n.github.io/convolutional-networks/>
- [30] A. Amidi and S. Amidi, *VIP Cheatsheet: Convolutional Neural Networks*. Accessed: Oct. 10, 2020. [Online]. Available: <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks>
- [31] X.-H. Le, G. Lee, K. Jung, H.-U. An, S. Lee, and Y. Jung, "Application of convolutional neural network for spatiotemporal bias correction of daily satellite-based precipitation," *Remote Sens.*, vol. 12, no. 17, p. 2731, Aug. 2020, doi: [10.3390/rs12172731](https://doi.org/10.3390/rs12172731).
- [32] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-k. Wong, and W.-c. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," 2015, *arXiv:1506.04214*. [Online]. Available: <http://arxiv.org/abs/1506.04214>
- [33] T. Mikolov, A. Joulin, S. Chopra, M. Mathieu, and M. Ranzato, "Learning longer memory in recurrent neural networks," 2014, *arXiv:1412.7753*. [Online]. Available: <http://arxiv.org/abs/1412.7753>
- [34] A. Graves, A.-R. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 6645–6649, doi: [10.1109/ICASSP.2013.6638947](https://doi.org/10.1109/ICASSP.2013.6638947).
- [35] C. Olah, *Understanding LSTM Networks*. Accessed: Jun. 6, 2020. [Online]. Available: <http://colah.github.io/posts/2015-08-Understanding-LSTMs/>
- [36] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*. [Online]. Available: <http://arxiv.org/abs/1412.3555>
- [37] M. Hermans and B. Schrauwen, "Training and analysing deep recurrent neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 26. Red Hook, NY, USA: Curran Associates, 2013, pp. 190–198. [Online]. Available: <https://proceedings.neurips.cc/paper/2013/file/1ff8a7b5dc7a7d1f0ed65aaa29c04b1e-Paper.pdf>
- [38] R. Pascanu, C. Gulcehre, K. Cho, and Y. Bengio, "How to construct deep recurrent neural networks," 2013, *arXiv:1312.6026*. [Online]. Available: <http://arxiv.org/abs/1312.6026>
- [39] H. Salehinejad, S. Sankar, J. Barfett, E. Colak, and S. Valaee, "Recent advances in recurrent neural networks," 2018, *arXiv:1801.01078*. [Online]. Available: <https://arxiv.org/abs/1801.01078>
- [40] J. E. Nash and J. V. Sutcliffe, "River flow forecasting through conceptual models part I—A discussion of principles," *J. Hydrol. Eng.*, vol. 10, no. 3, pp. 282–290, Apr. 1970, doi: [10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6).
- [41] G. Rossum, *Python Tutorial*. Amsterdam, The Netherlands: CWI (Centre for Mathematics and Computer Science), 1995.
- [42] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous distributed systems," 2016, *arXiv:1603.04467*. [Online]. Available: <http://arxiv.org/abs/1603.04467>
- [43] F. Chollet et al. *Keras*. GitHub, San Francisco, CA, USA. Accessed: Jun. 6, 2019. [Online]. Available: <https://github.com/fchollet/keras>
- [44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: <http://arxiv.org/abs/1412.6980>
- [45] S. Ruder, *An Overview of Gradient Descent Optimization Algorithms*. Accessed: Jun. 6, 2020. [Online]. Available: <https://ruder.io/optimizing-gradient-descent/>
- [46] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014, doi: [10.5555/2627435.2670313](https://doi.org/10.5555/2627435.2670313).



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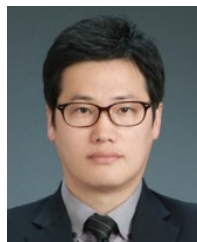


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